

Milestone 1: Project Definition Document

Project Title: Evaluation of Neuroevolutionary Control Strategies for
Autonomous Driving in Complex Road Topologies

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1. Problem Statement

The validation of Autonomous Vehicle (AV) control systems remains one of the most critical challenges in the automotive industry. While Deep Learning—specifically End-to-End Learning—has shown promise in mapping sensory inputs directly to control outputs, these systems often function as "black boxes," making their safety verification difficult. Furthermore, training these models typically requires vast amounts of labeled human driving data (Imitation Learning) or computationally expensive gradient-based Reinforcement Learning (RL) pipelines.

A distinct alternative is **Neuroevolution**, which optimizes neural weights using evolutionary algorithms (Genetic Algorithms) rather than gradient descent. This approach does not require labeled datasets and can potentially navigate non-differentiable fitness landscapes more robustly. However, a significant gap exists in understanding how these evolved agents perform in **structurally complex, real-world road networks** compared to traditional engineering solutions.

The core problem this project addresses is the lack of transparent, comparative performance data between Evolved Neural Networks and Rule-Based Heuristics in non-deterministic environments. Most current benchmarks rely on simple closed tracks or opaque commercial simulators (e.g., CARLA), which can obscure the underlying kinematic logic.

This project aims to:

1. Develop a transparent, custom-built 2D simulation environment capable of procedural and real-world (OpenStreetMap) map generation.
2. Implement and train a Neural Network-based agent using a Genetic Algorithm (Neuroevolution).
3. Quantitatively compare this agent against a Rule-Based Heuristic baseline (PID-style controller) across metrics of safety, efficiency, and trajectory smoothness.

2. Scope Definition

In accordance with the Capstone guidelines, the project scope is defined by the following choices:

- **Problem Type: Control & Optimization.** The system must output continuous control signals (steering, acceleration) to maximize an objective function (survival time, distance).
- **Data Type: Graph & Sensor Data.** The environment is constructed from geospatial graph data (Nodes/Ways from OSM), and the agent processes synthetic sensor data (Ray-Casting distances).
- **Technique Category: Neuroevolution (Deep Learning + Genetic Algorithms).** The primary method involves evolving the weights of a Feedforward Neural Network using selection, crossover, and mutation operators.
- **System Context: Comparative Analysis.** The final artifact is a simulation testbed designed to generate empirical data comparing two distinct control methodologies.

3. Data Source(s)

The project relies on two primary data sources:

1. **Synthetic Sensory Data (Internal):** The simulation will generate real-time LIDAR-like data via **Ray Casting**. The agent will receive normalized distance vectors representing the proximity of road borders and traffic obstacles.
2. **OpenStreetMap (OSM) (External):** To ensure the agent is tested on realistic topologies rather than just simple loops, the system will ingest real-world geospatial data.
 - **Source:** OpenStreetMap API / Overpass Turbo.
 - **Features:** Extraction of **node** (coordinates) and **way** (road segments) tags to construct the drivable spatial graph.
 - **Usage:** These data points will be parsed to generate road boundaries, intersections, and lanes within the custom physics engine.

4. Intended Methods

The project will be implemented using a custom JavaScript-based simulation stack (Canvas API) to ensure full control over the physics and rendering pipeline.

A. Environment & Physics Engine

- **Kinematics:** Implementation of a 2D rigid-body physics model including velocity, acceleration, friction, and collision detection (separating axis theorem or segment intersection).
- **Spatial Graph System:** A graph-based editor will be built to handle road segments, defining geometry for lane markings and borders.

B. Agent Architectures (The Comparison)

1. **The Neuroevolutionary Agent (Experimental):**
 - **Architecture:** A dense Feedforward Neural Network (Multi-Layer Perceptron).
 - **Training:** A Genetic Algorithm will manage a population of N agents.
 - *Fitness Function:* Defined by distance traveled vs. average speed.
 - *Operations:* Roulette selection, uniform crossover, and Gaussian mutation of weights.
2. **The Rule-Based Baseline (Control):**
 - **Logic:** A heuristic controller (Finite State Machine or PID) that strictly follows geometric rules (e.g., "if left sensor < threshold, steer right").

C. Evaluation Pipeline

- Both agents will be subjected to identical traffic scenarios.
- **Metrics:** Collision rate, Mean Distance Between Failures (MDBF), and "Jerk" (rate of change of acceleration) to measure driving smoothness.

5. Expected Outputs

1. **Simulation Software:** A functional, browser-based simulation environment capable of loading OSM data and training agents in real-time.

2. **Trained Models:** A serialized file (JSON) representing the optimized weights of the best-performing Neural Network.
3. **Comparative Report:** A detailed analysis contrasting the "learning curve" of the AI against the static performance of the heuristic baseline, specifically analyzing failure modes in complex intersections versus straight roads.
4. **Visualizations:** Real-time visualization of the Neural Network's activation states and sensor rays during the simulation.

6. Key Risks and Assumptions

Risk Category	Description	Mitigation Strategy
Convergence Failure	The Genetic Algorithm may fail to find a viable policy for complex OSM maps, resulting in agents that spin or stall.	Curriculum Learning: Train agents on simple tracks first, then transfer weights to complex maps. Implement "Elitism" to preserve best agents.
Computational Load	Calculating ray-cast intersections against thousands of OSM road segments may drop the frame rate, slowing training.	Spatial Partitioning: Implement a Quadtree or spatial hashing system to limit collision checks to local road segments only.
Sim-to-Real Gap	The 2D physics may not perfectly represent real vehicle dynamics (e.g., tire slip), limiting real-world applicability.	Assumption Acknowledgement: The project scope is explicitly bounded to <i>algorithmic evaluation</i> in simulation, not hardware deployment. This is a standard academic constraint.
Data Quality	OSM data often contains disconnected nodes or missing tags (e.g., lack of lane counts).	Graph Preprocessing: Implement a data cleaning module to "sew" disconnected graph edges and assign default widths to undefined roads.